

GateDrive

File: GateDrive.kicad_sch

uC

File: uC.kicad_sch

ExtPort

File: ExtPort.kicad_sch

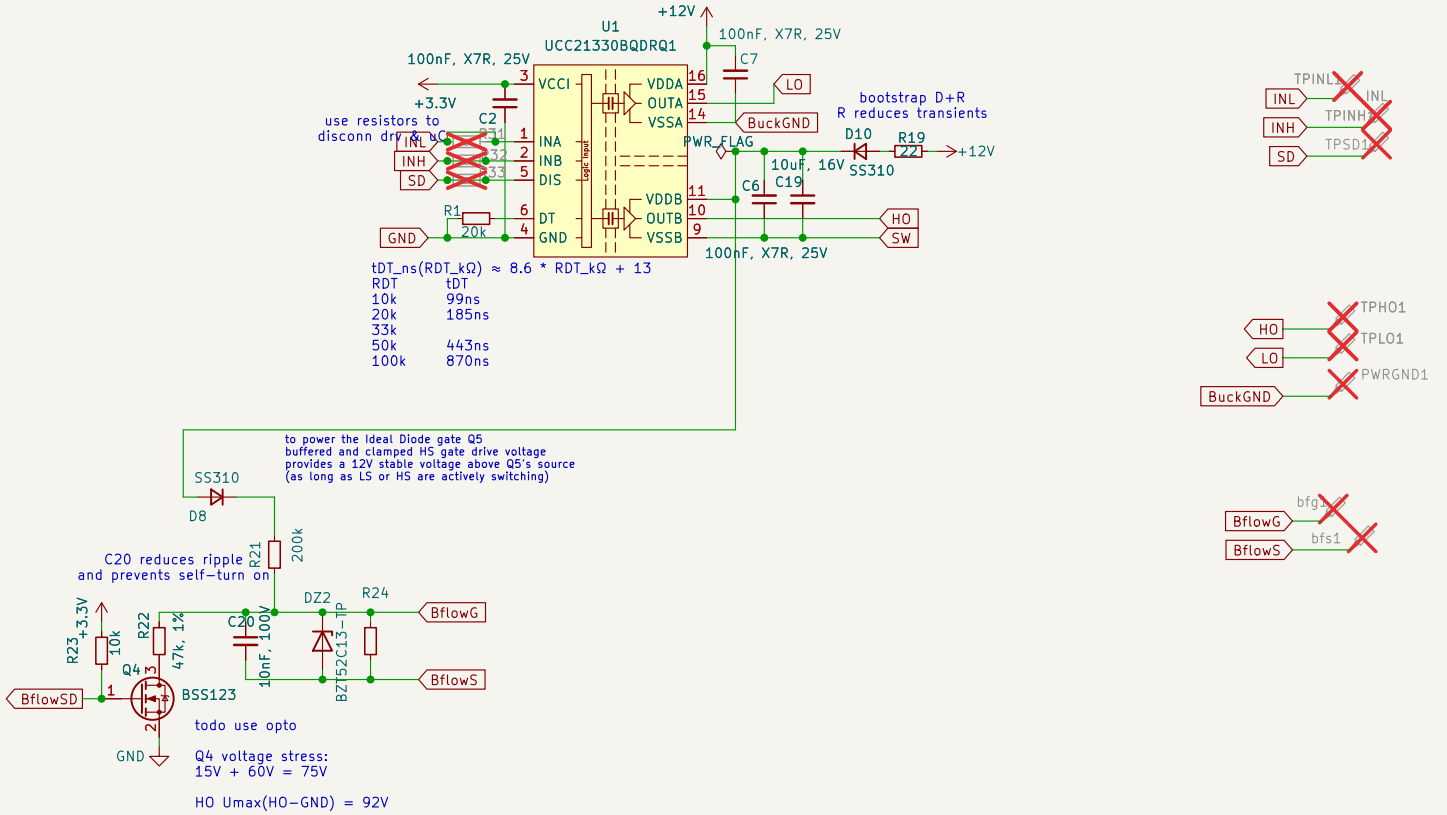
Sheetname: PowerSupply

TODO
- use 2p fuses?
- extension header 12V, 3.3V, i2c, io pins, SPI?
  -> UEXT connector (SPI, i2c, UART)
- LCD and buttons?
- replace Cin 10u with 4.7u
- external temp

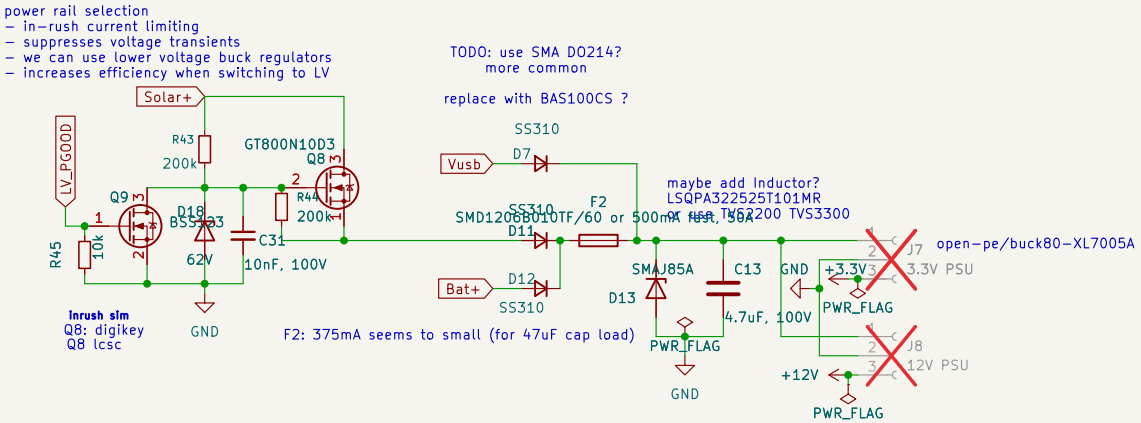
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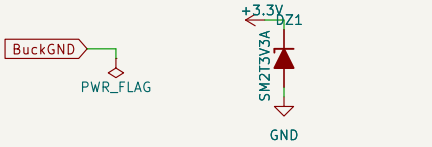
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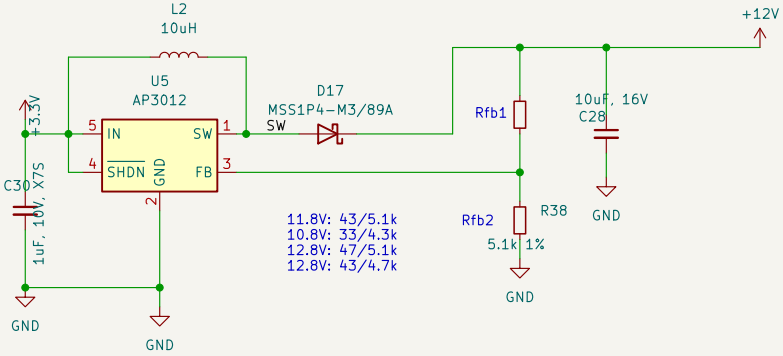
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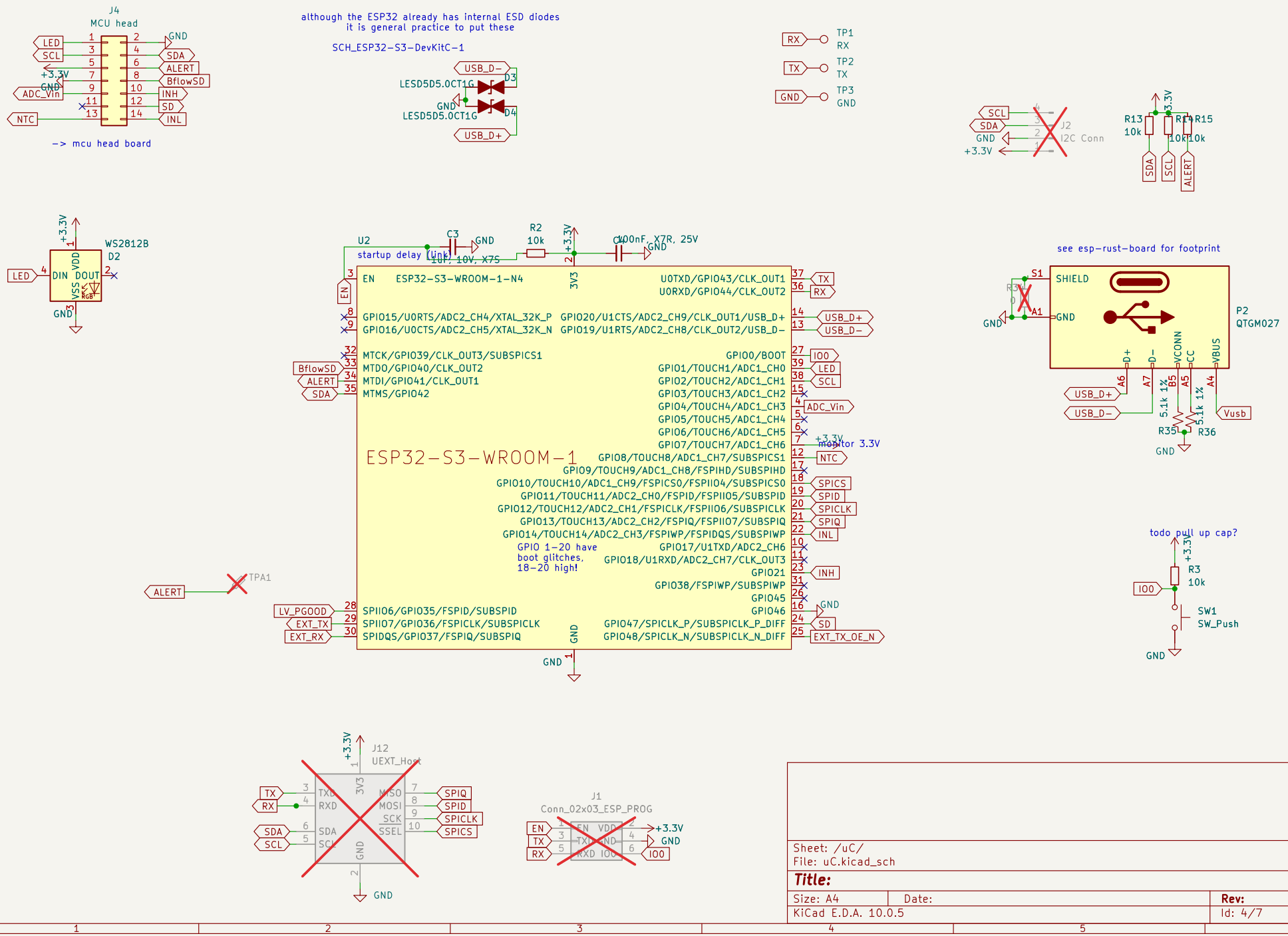
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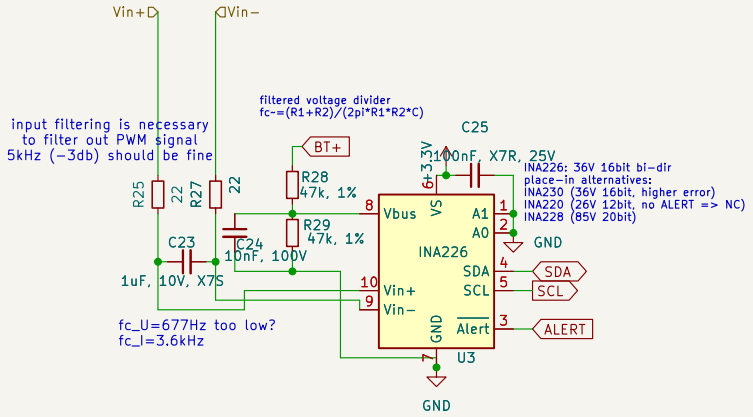


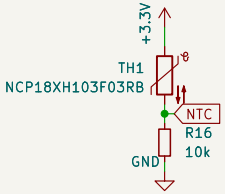


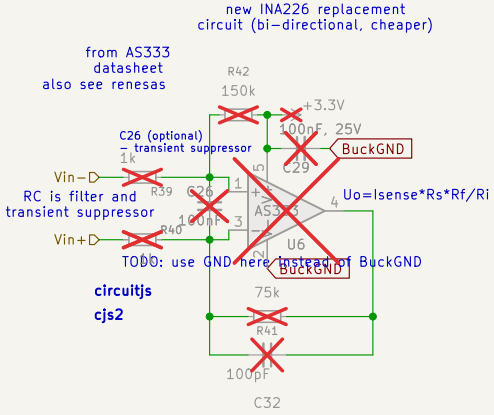
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J4
MCU head
LED
SCL
+3.3V
GND
ADC_Vin
NTC
GND
SDA
ALERT
BflowSD
INH
SD
INL
-> mcu head board
although the ESP32 already has internal ESD diodes
it is general practice to put these
SCH_ESP32-S3-DevKitC-1
USB_D-
LESD5D5.0CT1G
D3
GND
LESD5D5.0CT1G
D4
USB_D+
RX
TX
GND
TP1
RX
TP2
TX
TP3
GND
SCL
SDA
GND
+3.3V
J2
I2C Conn
R13
10k
R14
10k
R15
10k
SDA
SCL
ALERT
WS2812B
D2
LED
DIN
DOUT
VDD
VSS
GND
U2
startup delay (link)
C3
1uF, 10V, X7S
GND
R2
10k
+3.3V
C4
100nF, X7R, 25V
GND
EN
ESP32-S3-WROOM-1-N4
3V3
GPIO15/U0RTS/ADC2_CH4/XTAL_32K_P
GPIO16/U0CTS/ADC2_CH5/XTAL_32K_N
MTCK/GPIO39/CLK_OUT3/SUBSPICS1
MTDO/GPIO40/CLK_OUT2
MTDI/GPIO41/CLK_OUT1
MTMS/GPIO42
U0TXD/GPIO43/CLK_OUT1
U0RXD/GPIO44/CLK_OUT2
GPIO20/U1CTS/ADC2_CH9/CLK_OUT1/USB_D+
GPIO19/U1RTS/ADC2_CH8/CLK_OUT2/USB_D-
GPIO0/BOOT
GPIO1/TOUCH1/ADC1_CH0
GPIO2/TOUCH2/ADC1_CH1
GPIO3/TOUCH3/ADC1_CH2
GPIO4/TOUCH4/ADC1_CH3
GPIO5/TOUCH5/ADC1_CH4
GPIO6/TOUCH6/ADC1_CH5
GPIO7/TOUCH7/ADC1_CH6
GPIO8/TOUCH8/ADC1_CH7/SUBSPICS1
GPIO9/TOUCH9/ADC1_CH8/FSPIHD/SUBSPIHD
GPIO10/TOUCH10/ADC1_CH9/FSPICS0/FSPIIO4/SUBSPICS0
GPIO11/TOUCH11/ADC2_CH0/FSPID/FSPIIO5/SUBSPID
GPIO12/TOUCH12/ADC2_CH1/FSPICLK/FSPIIO6/SUBSPICLK
GPIO13/TOUCH13/ADC2_CH2/FSPIQ/FSPIIO7/SUBSPIQ
GPIO14/TOUCH14/ADC2_CH3/FSPIWP/FSPIDQS/SUBSPIWP
GPIO 1-20 have
boot glitches.
18-20 high!
GPIO17/U1TXD/ADC2_CH6
GPIO18/U1RXD/ADC2_CH7/CLK_OUT3
GPIO21
GPIO38/FSPIWP/SUBSPIWP
GPIO45
GPIO46
GPIO47/SPICLK_P/SUBSPICLK_P_DIFF
GPIO48/SPICLK_N/SUBSPICLK_N_DIFF
SPIIO6/GPIO35/FSPID/SUBSPID
SPIIO7/GPIO36/FSPICLK/SUBSPICLK
SPIDQS/GPIO37/FSPIQ/SUBSPIQ
ESP32-S3-WROOM-1
TX
RX
USB_D+
USB_D-
IO0
LED
SCL
ADC_Vin
monitor 3.3V
NTC
SPICS
SPID
SPICLK
SPIQ
INL
INH
GND
SD
EXT_TX_OE_N
BflowSD
ALERT
SDA
LV_PGOOD
EXT_TX
EXT_RX
ALERT
TPA1
see esp-rust-board for footprint
SHIELD
GND
P2
QTGM027
D+
D-
VCONN
CC
VBUS
A6
A7
B5
A5
A4
USB_D+
USB_D-
5.1k 1%
5.1k 1%
R35
R36
Vusb
GND
todo pull up cap?
+3.3V
R3
10k
IO0
SW1
SW_Push
GND
J12
UEXT_Host
+3.3V
TX
RX
SDA
SCL
TXD
RXD
SDA
SCL
3V3
GND
MISO
MOSI
SCK
SSEL
SPIQ
SPID
SPICLK
SPICS
GND
J1
Conn_02x03_ESP_PROG
EN
TX
RX
EN
VDD
TXD
GND
RXD
IO0
+3.3V
GND
IO0
Sheet: /uC/
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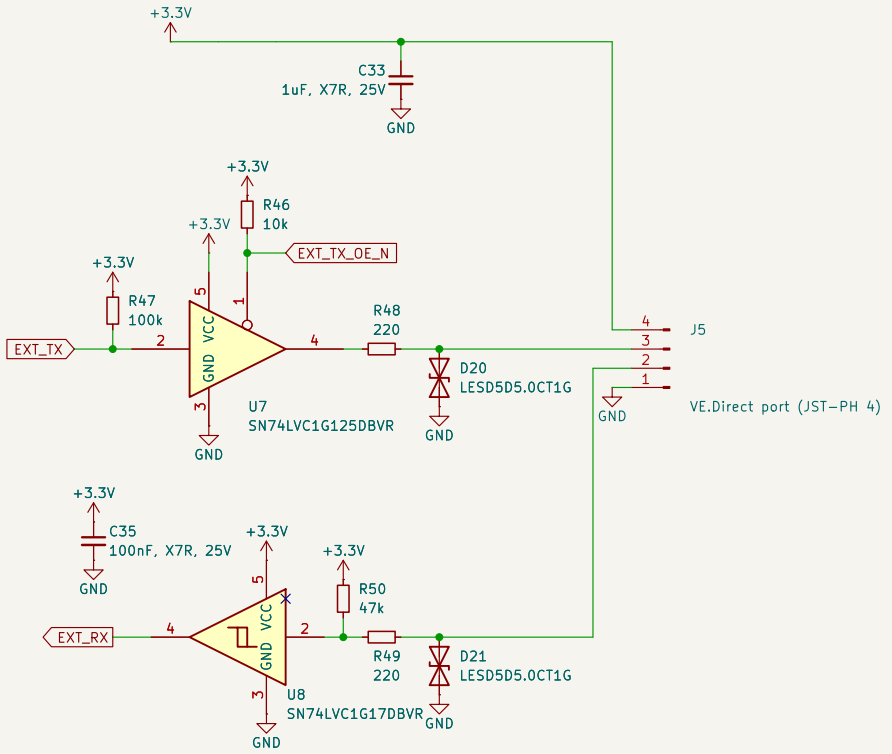
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Size: A4	Date:	**Rev:**
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External comms port, VE.Direct pinout (plans/ext-comms-port.md)

J5 pin | signal | dir (Fugu view) | notes
1 | GND | – | logic GND = Bat–; non-isolated
2 | RX | in | 3.3 V logic, 5.5 V tolerant (U8), idles high (R50)
3 | TX | out | 3.3 V push-pull via 220R, Hi-Z unless EXT_TX_OE_N low (R46 pull-up)
4 | V+ | out | +3.3V rail direct, always on (Fab, 2026-09-30). C33 at pin 4 = ESD sink + U7 bypass

MCU: EXT_TX GPIO36, EXT_RX GPIO37, EXT_TX_OE_N GPIO48
(not GPIO38/39: driven low in USB-OTG download boot mode, Espressif schematic checklist)
Wrong-adapter protection is in software only (Fab, 2026-09-28/29): no V+ switch, no V+ current sense.
No short protection and no back-feed block on pin 4 (Fab, 2026-09-29/30): a short browns out the
shared 0.5 A LM5163 rail; a peer driving pin 4 drives the 3.3 V rail.
Adapters: non-isolated, bus pins rated +/-70 V class (60 V battery GND fault); label them 'Fugu only'.

External comms port (VE.Direct pinout)

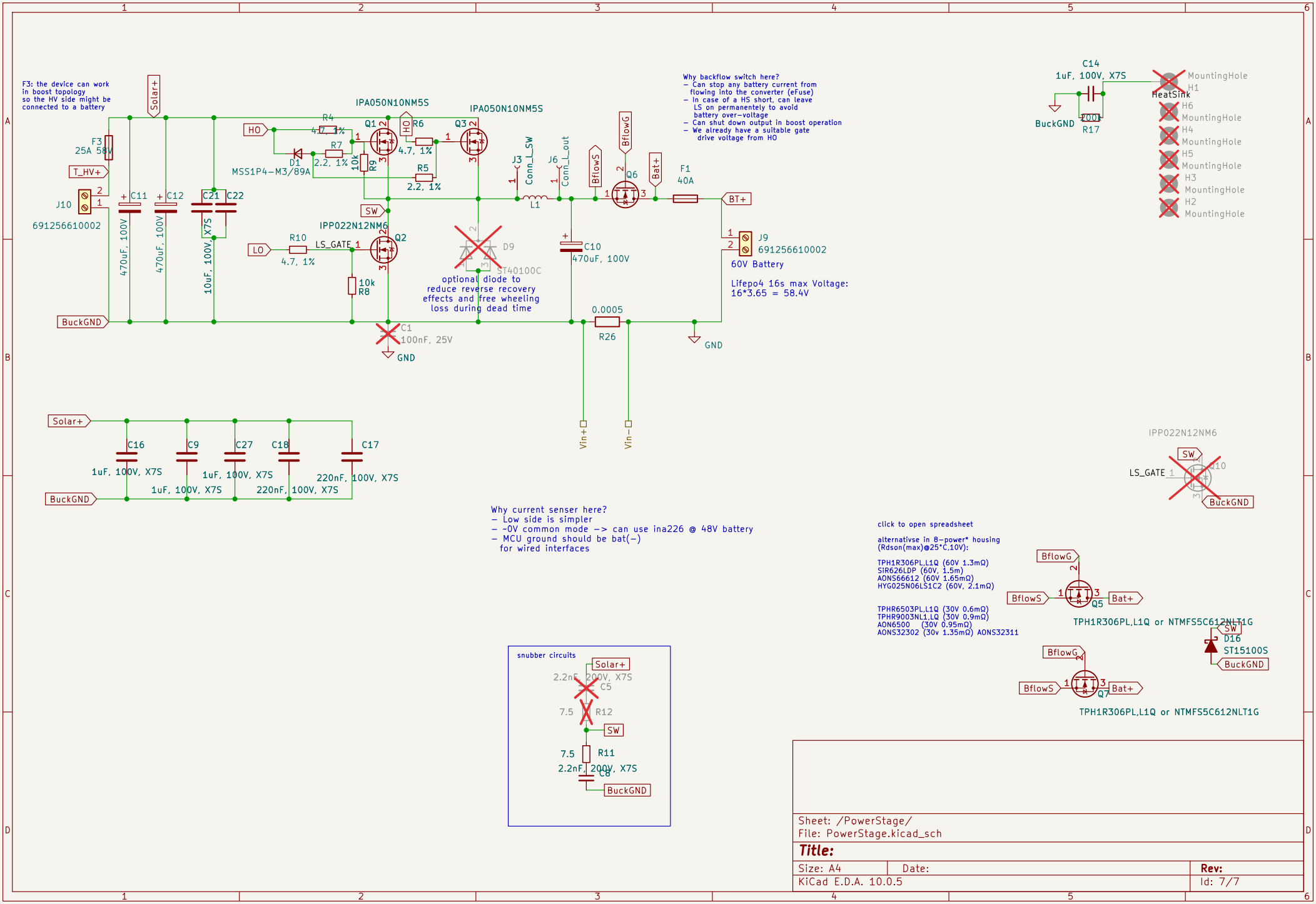


plans/ext-comms-port.md

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Title: External comms port

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F3: the device can work in boost topology so the HV side might be connected to a battery
Solar+
F3
25A 58V
T_HV+
J10
691256610002
C11
C12
C21
C22
470uF, 100V
470uF, 100V
10uF, 100V, X7S
BuckGND
IPA050N10NM5S
IPA050N10NM5S
HO
R4
4.7, 1%
Q1
HO
R6
4.7, 1%
Q3
R7
2.2, 1%
10k
R9
D1
MSS1P4-M3/89A
R5
2.2, 1%
SW
IPP022N12NM6
LO
R10
LS_GATE
4.7, 1%
Q2
10k
R8
D9
ST40100C
optional diode to reduce reverse recovery effects and free wheeling loss during dead time
C1
100nF, 25V
GND
J3
Conn_L_SW
L1
J6
Conn_L_out
BflowS
BflowG
Q6
Bat+
F1
40A
BT+
C10
470uF, 100V
0.0005
R26
J9
691256610002
60V Battery
Lifepo4 16s max Voltage:
16*3.65 = 58.4V
GND
Vin+
Vin-
Why backflow switch here?
– Can stop any battery current from flowing into the converter (eFuse)
– In case of a HS short, can leave LS on permanentely to avoid battery over-voltage
– Can shut down output in boost operation
– We already have a suitable gate drive voltage from HO
C14
1uF, 100V, X7S
BuckGND
200R
R17
HeatSink
MountingHole
H1
H6
H4
H5
H3
H2
Solar+
C16
C9
C27
C18
C17
1uF, 100V, X7S
1uF, 100V, X7S
220nF, 100V, X7S
220nF, 100V, X7S
BuckGND
Why current senser here?
– Low side is simpler
– ~0V common mode –> can use ina226 @ 48V battery
– MCU ground should be bat(–)
for wired interfaces
snubber circuits
Solar+
2.2nF, 200V, X7S
C5
7.5
R12
SW
7.5
R11
2.2nF, 200V, X7S
C8
BuckGND
IPP022N12NM6
SW
LS_GATE
Q10
BuckGND
click to open spreadsheet
alternatvse in 8–power* housing
(Rdson(max)@25°C,10V):
TPH1R306PL,L1Q (60V 1.3mΩ)
SiR626LDP (60V, 1.5m)
AONS66612 (60V 1.65mΩ)
HYG025N06LS1C2 (60V, 2.1mΩ)
TPHR6503PL,L1Q (30V 0.6mΩ)
TPHR9003NL1,LQ (30V 0.9mΩ)
AON6500 (30V 0.95mΩ)
AONS32302 (30v 1.35mΩ) AONS32311
BflowG
BflowS
Bat+
Q5
TPH1R306PL,L1Q or NTMFS5C612NLT1G
SW
D16
ST15100S
BuckGND
BflowG
BflowS
Bat+
Q7
TPH1R306PL,L1Q or NTMFS5C612NLT1G
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